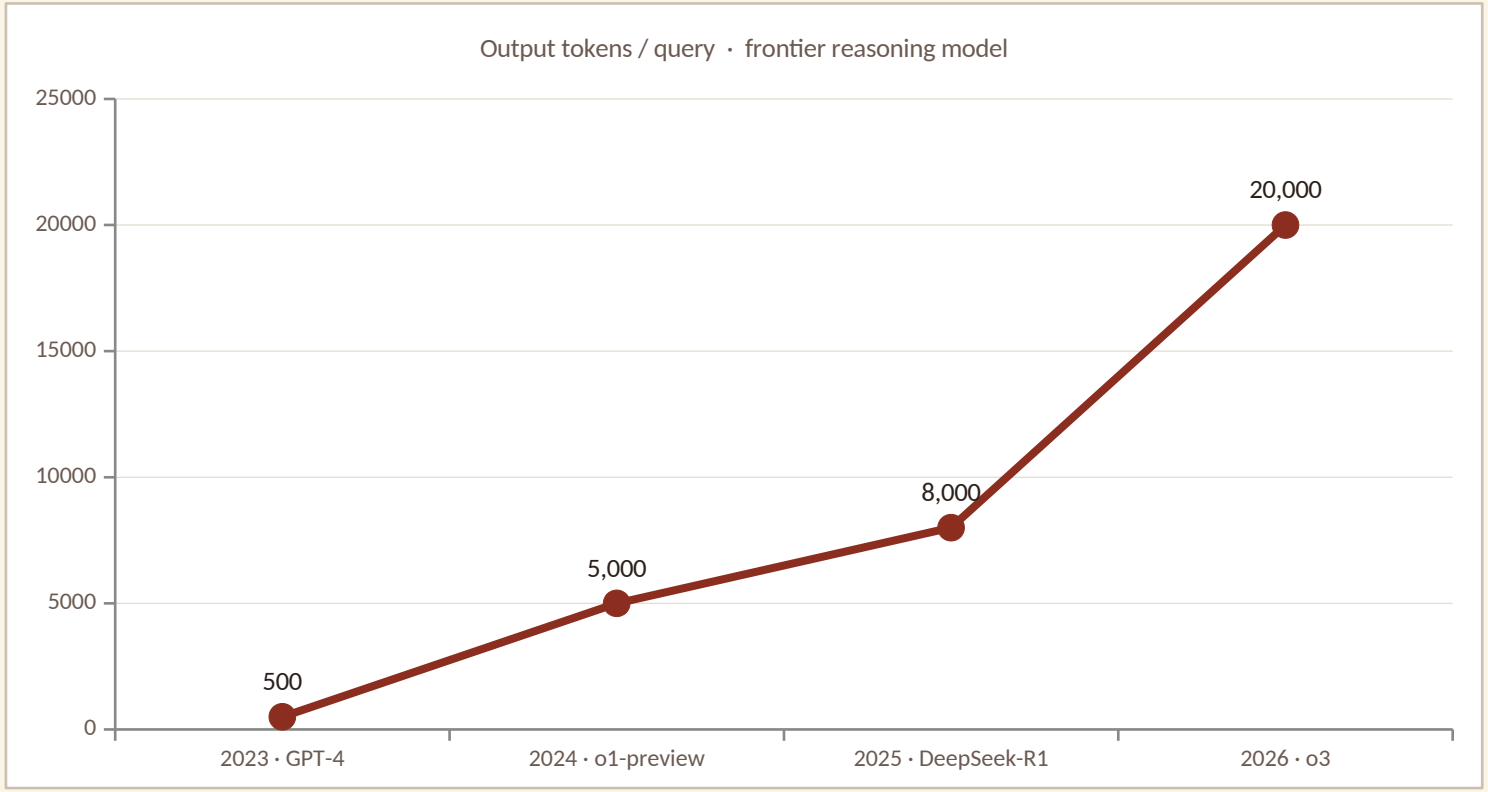
Reasoning burns tokens.
The substrate is the lever.



~40×

output-token growth per query, 24 months

Test-time compute is now a first-class scaling axis. The token substrate — shape and count — sets the constant in front of the cost curve.

Sam Larson, Founder & CEO · Jaime Barranca, President · sam@pebbleml.com

Byte-level is mainstream. Where's the lever?

BLT

Meta, Dec 2024

Meta discontinued discrete-token Chameleon and shipped BLT — byte-level, tokenizer-free. FAIR's bet on the substrate.

3

byte-level baselines, 2024–2025

EvaByte, MambaByte, MBLM — all tokenizer-free. Academic community converged on byte-level in 18 months.

~3.5×

more tokens per document

A byte carries ~1 nat. Subword tokens carry ~3–5. Byte-level means longer sequences, more test-time compute per query, lower info per token.

Byte-level removes the tokenizer but amplifies the underlying problem. The shape of each token is the open question. Pebble's bet is on one specific answer.

Matrix tokens: structure explicit per token.

Flat vector

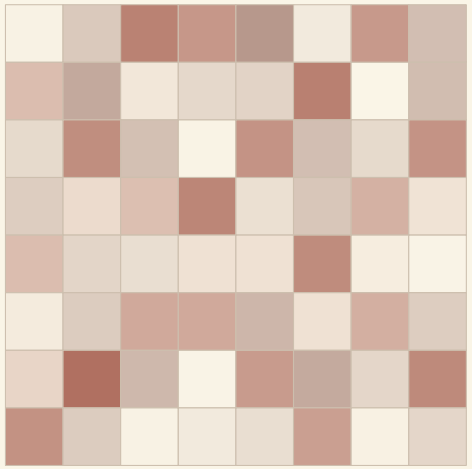
structure implicit across examples

```
[ 0.2, -0.8, 0.1, 0.6, ... ]
```

rank estimated across many inputs

Matrix token (d × d)

structure explicit · rank measured from one embedding



```
u ⊗ v  +  u' ⊗ v'  +  u'' ⊗ v''  + ...
```

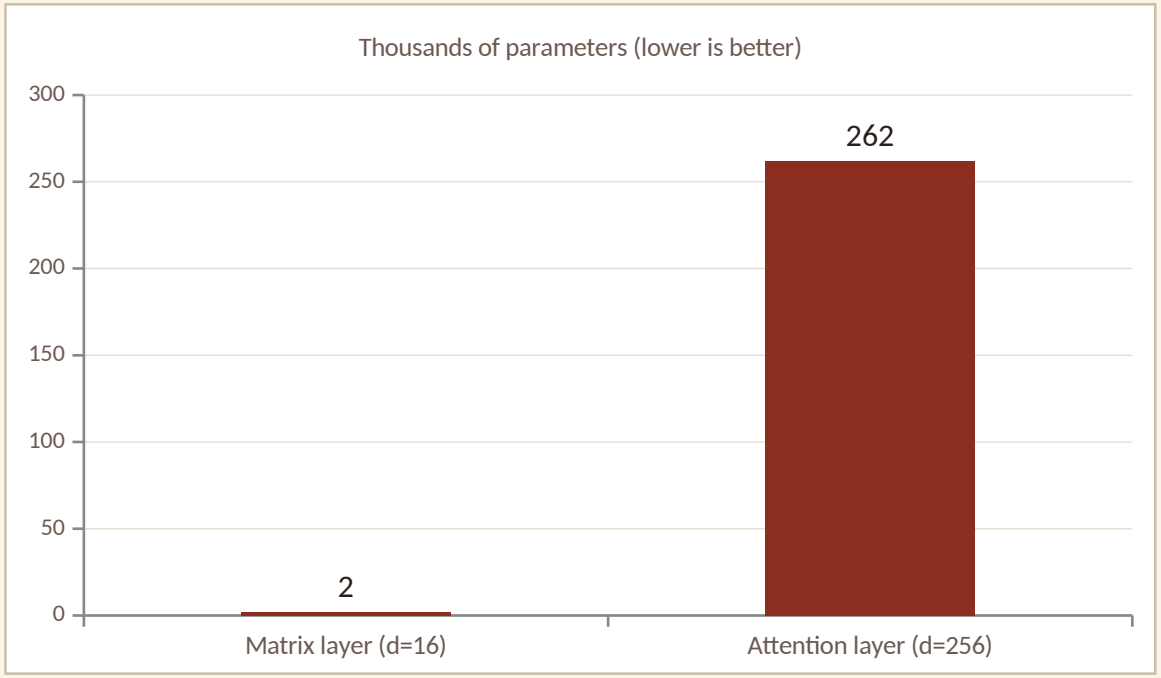
superposition of rank-1 outer products

Per-layer parameters, d=16 matrix layer vs d=256 attention layer: ~2K vs ~262K (~130× lower, structural). Working models use both matrix and non-matrix layers.

Per-layer efficiency is real. Quality signal is positive.

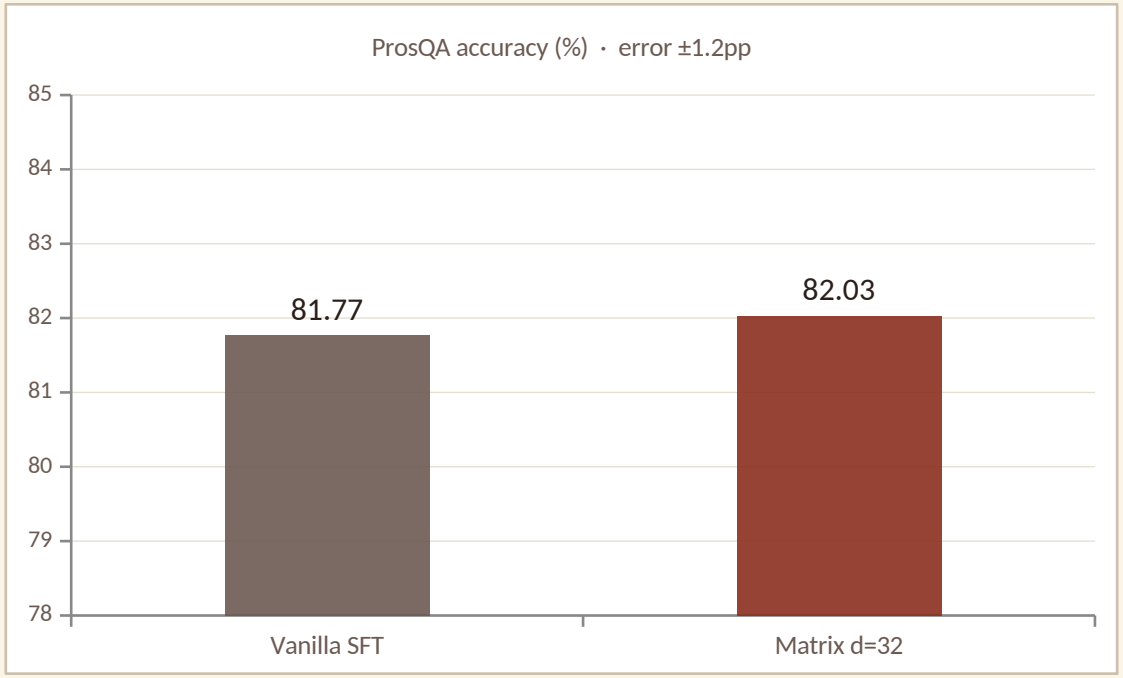
Params per layer at matched dimension

Matrix d=16 vs standard attention d=256. Structural.



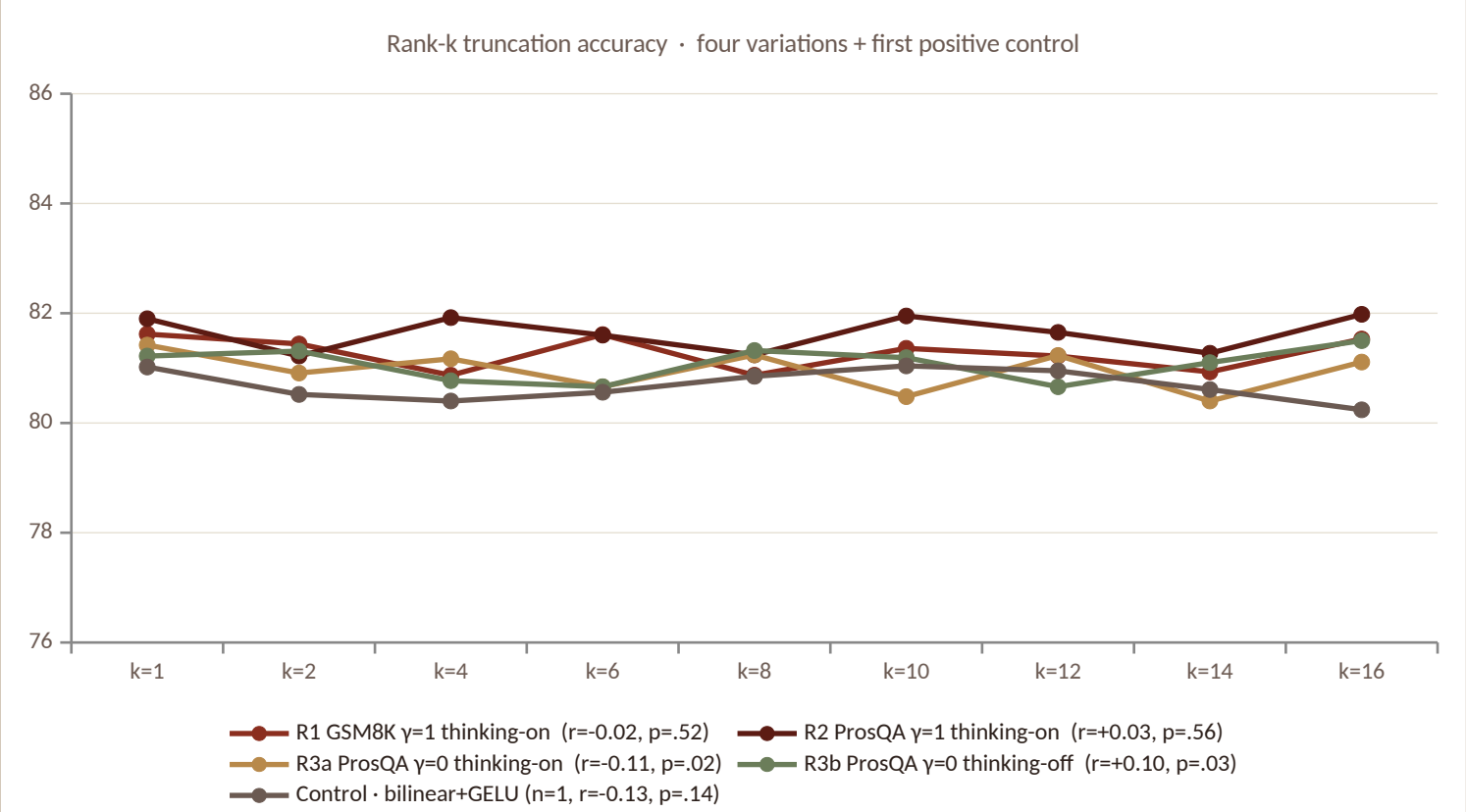
ProsQA accuracy (3-seed mean)

Reproducible positive direction, 3 seeds.



Per-layer efficiency is structural. The quality signal is early — the real test is a matrix-native architecture end-to-end. That's Chapter 2.

Bolt-on architecture hides the structure. We know why.



Spearman r

R1 GSM8K γ=1	**−0.023**	p=.519
R2 ProsQA γ=1	**+0.026**	p=.558
R3a ProsQA γ=0 on	**−0.105**	p=.018
R3b ProsQA γ=0 off	**+0.095**	p=.034

Control (n=1)

bilinear+GELU readout

r = −0.13 p = 0.14

Not significant at α=.05. First of five planned controls.

3-seed flatten replication

Accuracy 81.5 ± 1.2pp · Z_rank varies 3× (4, 12, 13)

The flatten readout destroys the gradient path through rank. Four flat rank-k curves; every nonlinear control we tried replicates the pattern. Fix: go matrix-native end-to-end.

Matrix-native architecture. This is where the upside lives.

EXPERIMENT SPEC

Matrix-native transformer on synthetic rank-K task

ARCHITECTURE

d×d matrix tokens throughout. No vector pretraining, no CODI distillation, no flatten readout. Matrix attention (Frobenius inner product), matrix MLP preserving row/column structure, MultiProbeHead output. ~5–10M params.

TASK

K-parallel entity tracking (Task A). Optimal solution provably rank-K. Task generator passes self-tests in chapter2/synthetic_tasks.py. Two additional tasks scoped.

DECISION

pre-registered

Survive if |Spearman r| > 0.3, p < 0.01, across 3 seeds, on ≥2 of 3 tasks. Fail if |r| < 0.15 across all tasks. Mixed → task-structure investigation. Fail → pivot to byte-level JEPA.

BUDGET

~75 H100-hours including ablations. Single decision point.

DECISION CRITERION

|r| > 0.3

p < 0.01

3 seeds

≥ 2 of 3 tasks

Clean yes / no.

Tightens CHAPTER_2_DESIGN.md.

Three concrete directions for productizing the research.

Byte-level multimodal.

The gap bGPT left open.

bGPT trained on text + images + audio from raw bytes. Negative transfer between text and non-text. Flat-vector bytes can't encode 2D spatial layout; matrix rows and columns can, without special position codes. Nobody has combined byte-level + matrix + multi-domain.

Efficient specialized models.

Per-layer efficiency, productized.

130× fewer parameters per matrix layer at matched width. The path: smaller models that reason better per token — deployable as APIs for domains where inference cost is the bottleneck. Code, math, interp tooling.

Rank as an interp primitive.

Measured, not estimated.

Rank is a structural property of a matrix token. Most interp programs estimate it across ensembles — we compute it per sample. A substrate-level observable that unlocks new probes, metrics, and experiments for labs that buy interpretability.

Same research platform, three revenue lines. All of it runs on the compute you're already amortizing.

The billion-param programs are betting on architecture. We're betting on substrate.

GROUP	LAYER OF THE BET	SCALE	POSITION
HELM (NeurIPS 2025)	Architecture (hyperbolic)	Billion-param	Scale proven. Published.
JEPA / AMI Labs	Objective (predictive SSL)	Billion-scale in flight	~$1B raised, Feb 2026.
Cartesia (Gu et al.)	Sequence model (SSM)	Production scale	VC-backed, commercial track.
CRATE (Yi Ma)	Model (white-box)	Research-scale	Academic program.
Pebble AI	**Substrate (matrix tokens)**	**288K — 355M → scaling**	**First published diagnostic on the substrate layer. ICML MI Workshop paper shipping.**

Everyone above picked a layer above the token. Pebble is one layer deeper. New infrastructure gets built at the substrate — CUDA, PyTorch, Transformers all won by being the right shape underneath. That's the position we're taking.

A research program that runs itself.

Autonomous agent harness. Grabs a GPU. Runs all day. Compounds learnings.

QUEUE
`agent-refilled`

→

AUDIT
`separate agent`

→

RUN
`any available GPU`

→

BANK
`results → SQLite`

→

COMPOUND
`rules auto-inject`

Preemption-recovering. Wakeup-chain scheduled. No human in the inner loop.

INTERNAL

Substrate research, in-house.

The fuel behind the workshop paper and Chapter 2.

- Queue candidates go through pre-experiment checklist (hypothesis, FLOPs, prior-art scan) before compute is spent.
- Learnings DB (SQLite + FTS5) auto-injects relevant prior rules into every new prompt. Mistakes stop repeating.
- Wakeup chain schedules next run before GPU would idle — no wasted minutes between experiments.

EXTERNAL

Research-as-a-service.

Same platform. Any ML research workload.

- Client defines the question; the harness runs the search across small-scale experiments on whatever GPU is cheapest.
- Audit-then-run pipeline is domain-agnostic — applies equally to a customer's architecture search as to our own.
- Goodfire-style research contracts are the near-term revenue line. Platform scales horizontally with GPU supply.

Most labs are humans at terminals. We built the research platform first — and it already ships papers.

Three revenue paths. Compute-aligned ask. Small team.

MONETIZATION

Research-as-a-service

`near-term`

Contract architecture-search and interp studies on the harness. Goodfire is the closest comparable.

Specialized capability APIs

`medium-term`

Package a winning Chapter 2/3 architecture as a narrow API — code, math, or interp-primitive workloads on cheaper inference.

Licensing + acquihire

`optionality floor`

Novel-architecture IP on the frontier labs' silicon. Adept → Amazon, Inflection → MS, Character → Google (2024) set the comps.

ASK

NVIDIA Inception

`~135 H100-hrs`

Compute across 12 months. Researcher-network access. DLI credits if available.

Startup credits

`$50 – 100K`

AWS / Google / Azure for storage, egress, non-H100 work.

Technical advisors

`1 – 3 warm intros`

PhD-level, reasoning or interp background.

TEAM

Sam Larson

Founder & CEO · research

Built the matrix-thinking research program and the autonomous harness behind it. Author of the ICML MI Workshop 2026 paper on substrate-layer diagnostics. Sets the research agenda and operates the lab.

Jaime Barranca

President · venture, strategy

Venture-capital background. Drives business strategy, capital formation, and the partner network that converts substrate research into infrastructure.

PEBBLE AI

A research program that compounds.

Built to find the next architecture.

Workshop paper May 8. Chapter 2 next. Machinery already running.

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